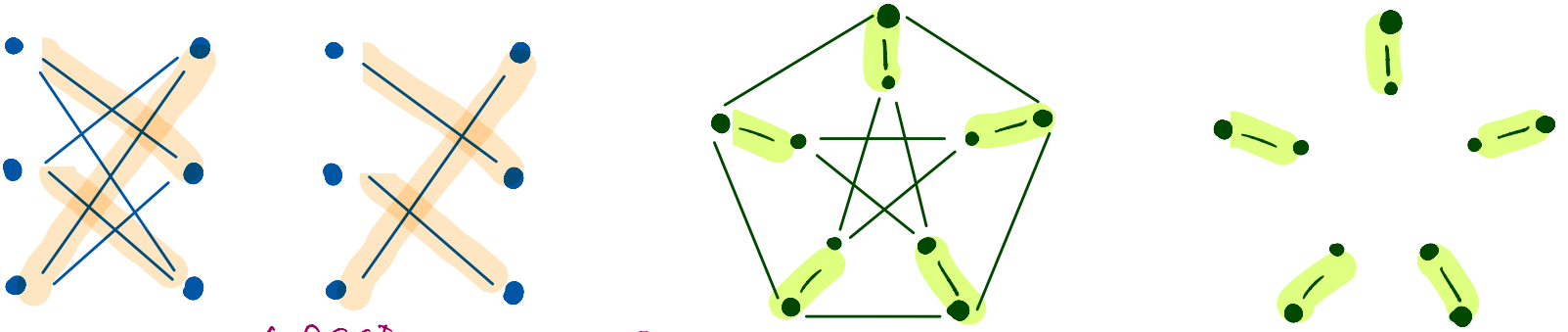
Recall: For a graph $G = (V, E)$, a perfect matching is a subset $M \subseteq E(G)$ such that each vertex is incident to exactly one edge in M.

Question. If M is a perfect matching, what type of graph is (V, M)?

Example.



Definition. *A* 1-***factor*** *of a graph* G *is a* 1-*regular spanning subgraph of* G.

Remark: Let $G = (V, E)$, and consider any subset $M \subseteq E$.
Then M is a perfect matching if and only if (V, M) is a 1-factor of G.

Definition. *A connected component of a graph G is called* ***odd*** *or* ***even*** *if it has an odd or even number of vertices, respectively. The number of odd components of G is denoted* $k_{odd}(G)$.

Theorem (Tutte, 8.10). *A graph G contains a perfect matching if and only if*

$$k_{odd}(G - S) \leqslant |S| \quad \textit{for every } S \subseteq V(G).$$

) (a.k.a. 1-factor)

Proof: $\Rightarrow$) Let M perfect matching. Let $S \subsetneq V(G)$

Let $G_1, \dots, G_K$ odd components of $G-S$ ($K = k_{odd}(G)$)

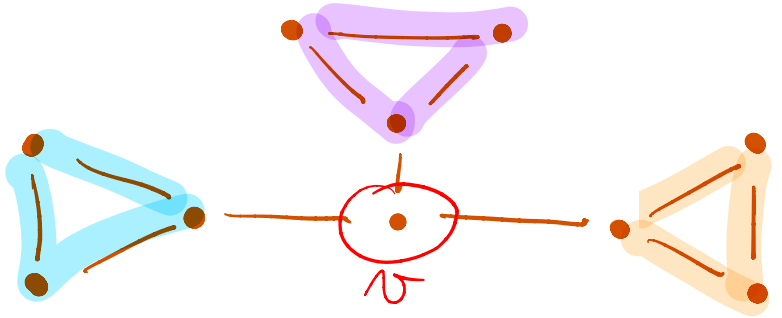
Claim: For each G_i, There are $v_i \in V(G_i)$ and $s_i \in S$ so That $v_i s_i \in M$.

All vertices in G_i are matched by M. But $|V(G_i)|$ is odd, and M matches an even number of vertices. Thus some $v_i \in V(G_i)$ is matched by M To some vertex not in $V(G_i)$. That vertex needs To be in S.

Moreover, if $i \neq j$, Then $s_i \neq s_j$ (M is a matching)

Foreach G_i There is an element in S, Thus $K \leq |S|$ ☆

Example. *Decide if the following graph contains a perfect matching.* No, $S = \{v\}$



1-factor

Definition. *A graph G is* **1-factorable** *if we can partition E(G) as a disjoint union of perfect matchings.*

Theorem (8.14). *The complete graph K_n for n even is 1-factorable.*

Proof: $E(K_n) = M_1 \cup \dots \cup M_k$ (perfect matchings)] Done last time!

Theorem (8.15). *Every regular bipartite graph is 1-factorable.*

Proof

Done: HW 10.3.

Last time: For a graph G,

- A proper edge-coloring is a function $f: E(G) \to C$ where $f(e) \neq f(s)$ if e, s have a common vertex.
- A proper edge coloring using t colors corresponds to a partition of $E(G)$ into t matchings.
- The edge chromatic number $\chi'(G)$ is the smallest number of colors in any proper edge-coloring.
- Vizing's Theorem: $\Delta(G) \leqslant \chi'(G) \leqslant \Delta(G) + 1$.

Next: Some conditions to decide whether $\chi'(G) = \Delta(G)$ or $\chi'(G) = \Delta(G) + 1$.

Theorem (10.13). *Let G be an n-vertex graph, where n is odd. If $|E(G)| > \frac{n-1}{2}\Delta(G)$, then $\chi'(G) = \Delta(G) + 1$.*

Proof: For any proper edge coloring, no more than $\frac{n-1}{2}$ edges can be colored the same ($\frac{n-1}{2}+1$ edges will have a common vertex, otherwise we have $2\left(\frac{n-1}{2}+1\right) = n+1$ distinct vertices). But then with $\Delta(G)$ colors, we can color at most $\frac{n-1}{2}\Delta(G)$ edges. But we have $|E(G)| > \frac{n-1}{2}\Delta(G)$. Thus $\chi'(G) > \Delta(G)$. ∎

Theorem. *Let* G *be a* r*-regular graph. Then* $\chi'(G) = \Delta(G)$ *if and only if* G *is* 1*-factorable.*

$E(G) = M_1 \cup \cdots \cup M_k$ perfect matchings.

Proof: HW 12

1) $\Delta(G) = r$

$\Rightarrow$) $E(G) = M_1 \cup \cdots \cup M_r$ matchings

2) If M is matching, Then $|M| \leq \frac{n}{2}$ $(n = |V(G)|)$

With = iff M is perfect.

3) $|E(G)| = \frac{1}{2} nr$.

$\Leftarrow$) $E(G) = M_1 \cup \cdots \cup M_k$ → Can be $k > r$?

Theorem (König, 10.17). *If* G *is a bipartite graph, then* $\chi'(G) = \Delta(G)$.

Let $r = \Delta(G)$, Let U_0, W_0 denote the parts of G

Lemma: If G is not regular, then $G \subseteq G_1$ for some G_1 so that

1) G_1 bipartite, 2) $\delta(G_1) = \delta(G) + 1$, 3) $\Delta(G_1) = r$

Proof: Let G' denote a copy of G, with parts U_0', W_0'

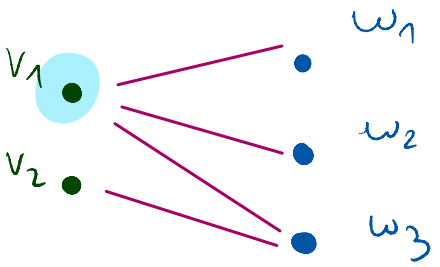
Let G_1 obtained from G and G' by joining each $v \in V(G)$ with $\deg_G v < r$, with the corresponding $v' \in V(G')$

1) The parts are $U_0 \cup W_0'$ and $W_0 \cup U_0'$

Need to show: e.g. no edge in G_1 goes from U_0 to W_0'

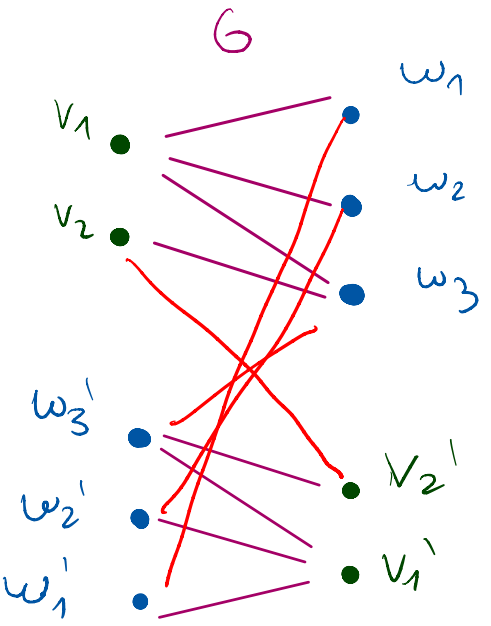
edges in G go from U_0 to W_0 / new edges go from G' go from W_0' to U_0' / U_0 to U_0'

Example:



G

$r = \Delta(G) = 3$



G

G'

G_1

2) $\delta(G_1) = \delta(G) + 1$

Each $v \in V(G)$ has $\deg_{G_1} v = \begin{cases} \deg_G v & \text{if } \deg_G v = r \\ \deg_G v + 1 & \text{otherwise} \end{cases}$

↓

add new neighbor v' in G'

3) $\Delta(G_1) = \Delta(G)$: No new neighbors for vertices of maximum degree r. ▨ Lemma

Consequence: If G is not regular, let $k = \Delta(G) - \delta(G)$

Apply Lemma k-Times to G, get a graph $G_k \supseteq G$ and

1) G_k bipartite, 2) $\delta(G_k) = \delta(G) + k$, 3) $\Delta(G_k) = \Delta(G) = r$
$= r$

ie: G_k is r-regular.

G^{ν}

By Thm above, G_k is 1-factorable

$$E(G_k) = M_1 \cup \cdots \cup M_r$$

($M_1, \ldots, M_r$: perfect matchings; r because of previous Thm)

Then

$$E(G) = (E(G) \cap M_1) \cup \cdots \cup (E(G) \cap M_r)$$

(each $E(G) \cap M_i$: matchings.)

This gives a proper edge coloring for G using r colors.

Hence $\chi'(G) \leq r = \Delta(G)$. $\quad (\chi'(G) \geq \Delta(G))$

By Vizing's, $\chi'(G) = \Delta(G)$. ∎